HIGH PERFORMANCE RESEARCH COMPUTING

Using a Passenger App for Displaying GPU Node Configuration and Resource Availability

Michael Dickens

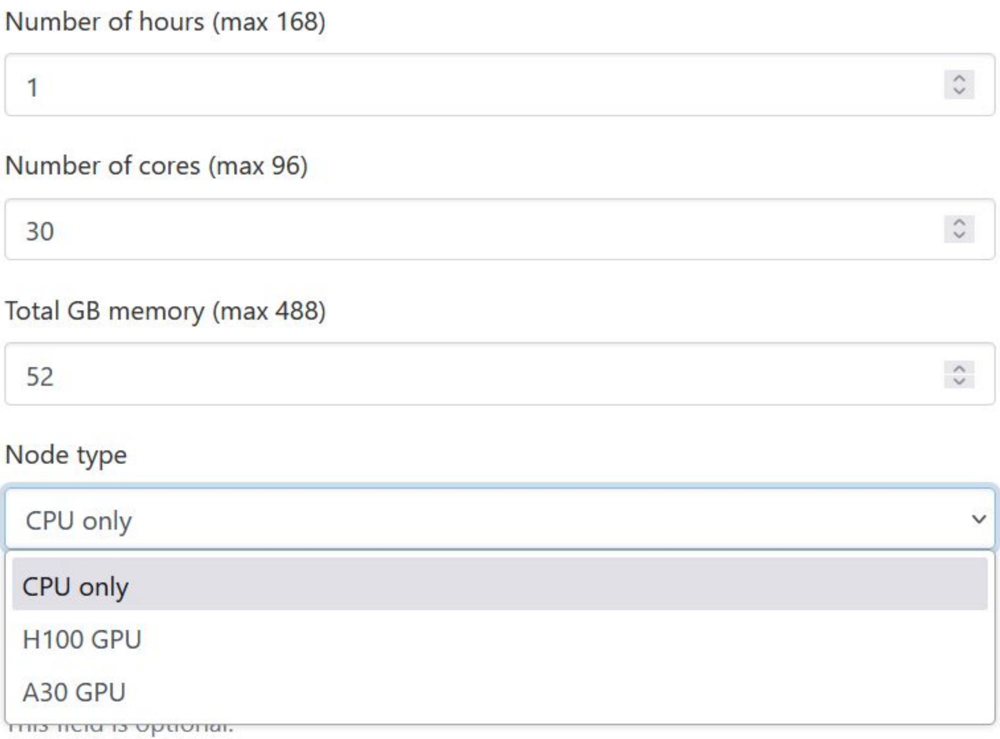
Texas A&M University

GOOD

Global Open OnDemand Conference

HARVARD UNIVERSITY | MARCH 18–20, 2025

Selecting GPUs for Interactive Apps



Questions when Selecting GPUs for Interactive Apps

CONFIGURATION

- What are the current GPUs configurations on the cluster
  - how many of each type of GPU are installed per node
  - how many nodes of each GPU configuration are installed

AVAILABILITY

- How many GPUs, CPUs and GB of memory per node are currently available for interactive jobs

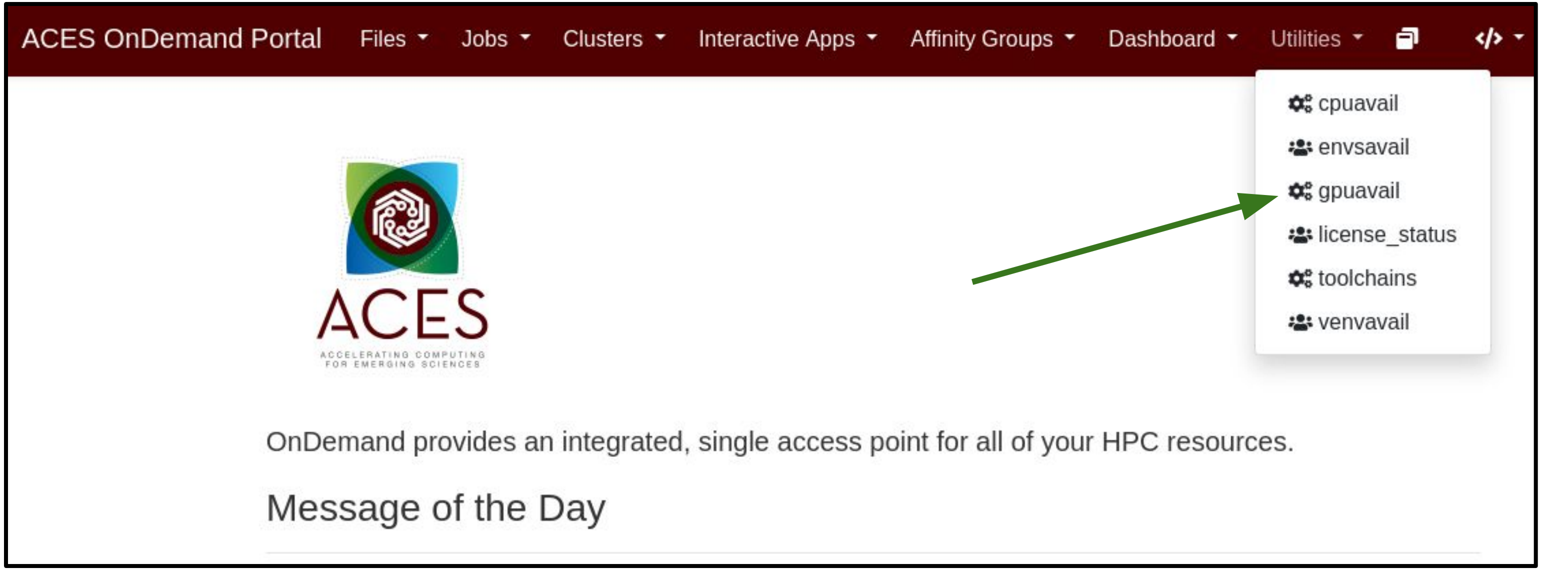
Current GPU node Configurations on HPRC Clusters

ACES*+	FASTER*+	Grace	Launch+
gpu:h100:2	gpu:a100:4	gpu:a100:2	gpu:a30:2
gpu:h100:4	gpu:a100:5	gpu:a40:3	
gpu:h100:8	gpu:a100:6	gpu:rtx:2	
gpu:a30:2	gpu:a100:8	gpu:t4:4	
gpu:pvc:2	gpu:a100:16		
gpu:pvc:4	gpu:t4:4		
gpu:pvc:8	gpu:t4:8		
	gpu:a40:2		
	gpu:a40:4		
	gpu:a30:2		
	gpu:a10:2		
	gpu:a10:4		

* Hardware composable clusters

+ available through ACCESS https://access-ci.org

gpuavail Accessible as a Utility



ACES OnDemand Portal / ACES GPU Availability

ACES GPU Availability

This displays the CONFIGURATION of installed GPUs and the status of GPUs readily available for jobs.

GPUs and GPU nodes not in the AVAILABILITY table are busy with other jobs and will be available after jobs have completed.

Output generated: **2025-03-15 20:00:14**

```
$ gpuavail
```

CONFIGURATION	
NODE TYPE	NODE COUNT
gpu:pvc:4	17
gpu:pvc:2	11
gpu:pvc:8	4
gpu:h100:4	3
gpu:h100:8	2
gpu:h100:2	1
gpu:a30:2	1

AVAILABILITY					
NODE NAME	GPU TYPE	GPU COUNT	GPU AVAIL	CPU AVAIL	GB MEM AVAIL
ac064	a30	2	**1**	88	456
ac013	pvc	4	**4**	96	488
ac024	pvc	8	**8**	96	488
ac100	pvc	2	**2**	96	488
ac101	pvc	4	**4**	96	488
ac082	pvc	2	**2**	96	488
ac085	pvc	4	**4**	96	488
ac099	pvc	4	**4**	96	488
ac103	pvc	2	**2**	96	488

ACES OnDemand Portal / ACES GPU Availability

ACES GPU Availability

This displays the CONFIGURATION of installed GPUs and the status of GPUs readily available for jobs.

GPUs and GPU nodes not in the AVAILABILITY table are busy with other jobs and will be available after jobs have completed.

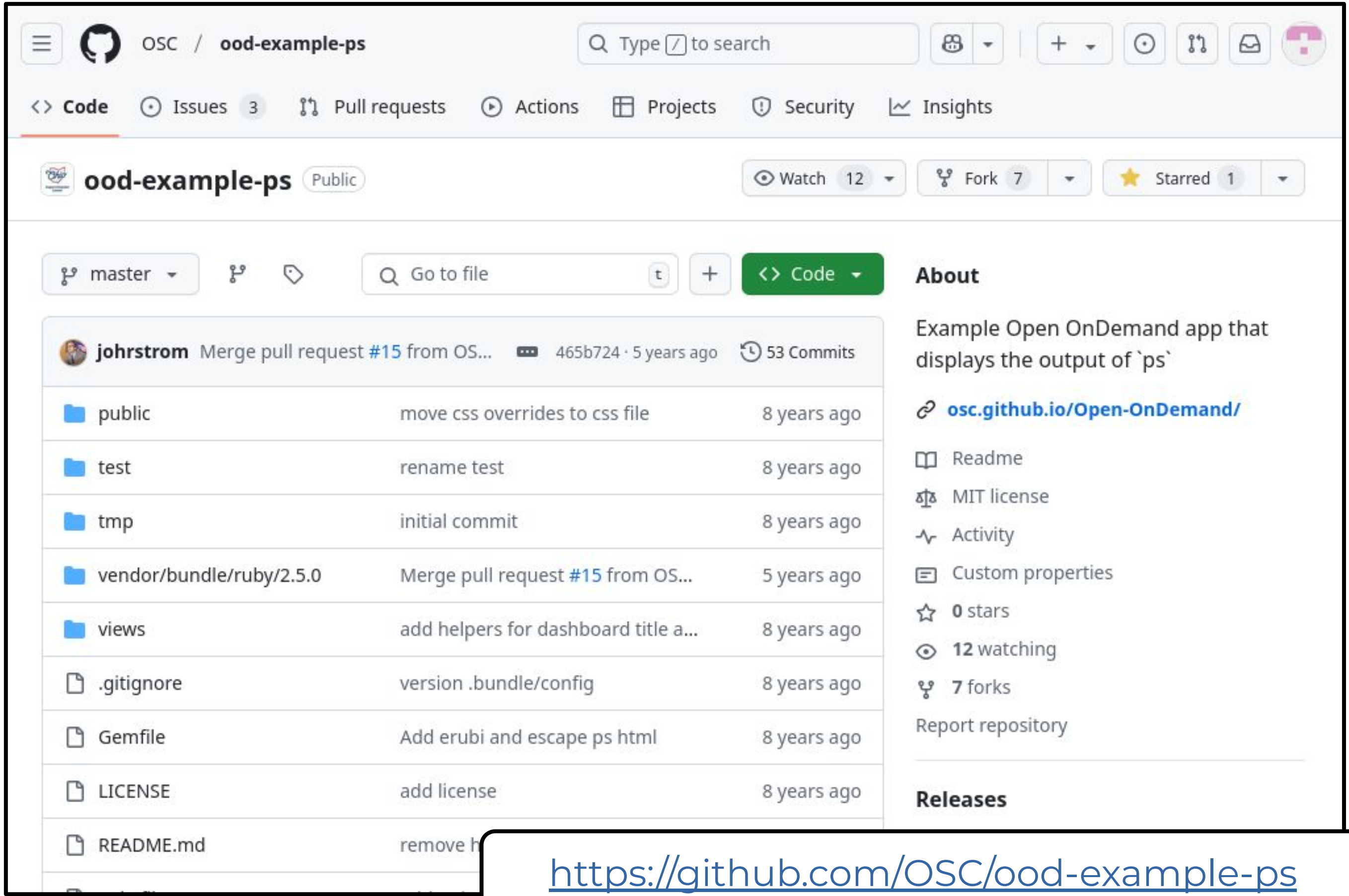
Output generated: **2025-03-15 20:00:14**

```
$ gpuavail
```

CONFIGURATION	
NODE TYPE	NODE COUNT
gpu:pvc:4	17
gpu:pvc:2	11
gpu:pvc:8	4
gpu:h100:4	3
gpu:h100:8	2
gpu:h100:2	1
gpu:a30:2	1

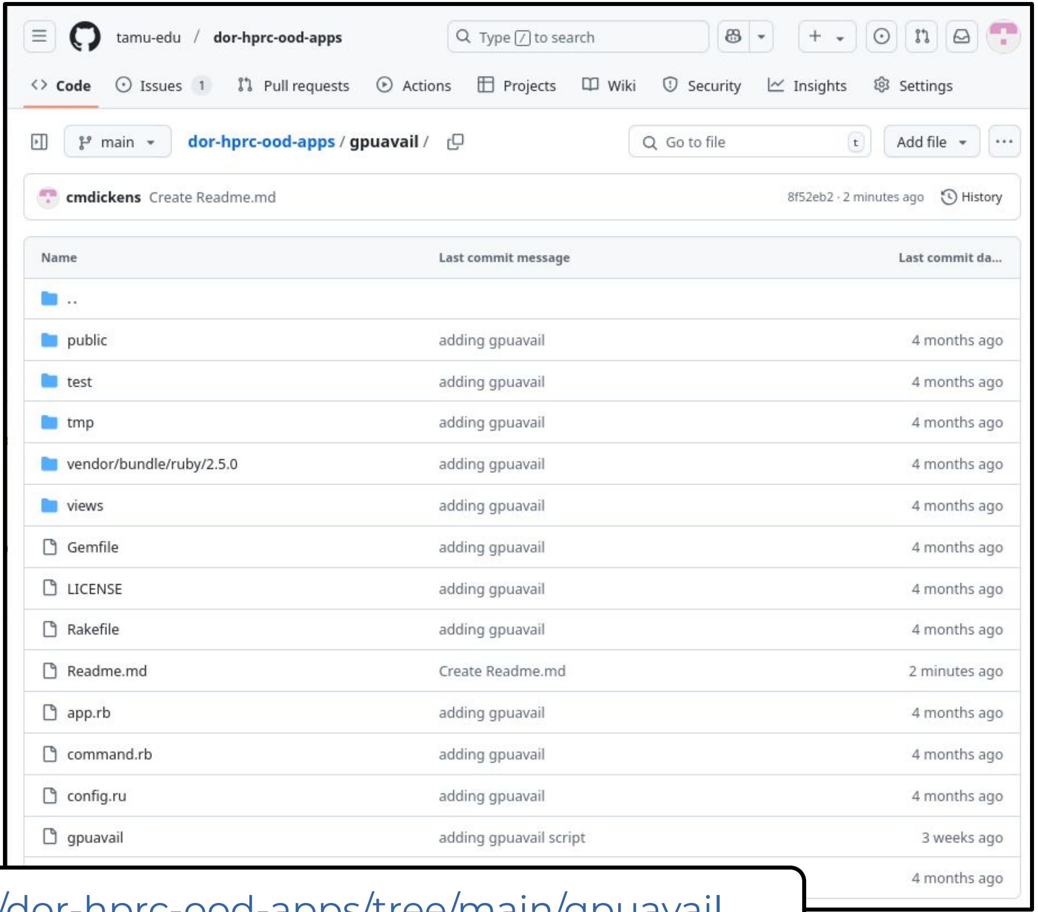
AVAILABILITY					
NODE NAME	GPU TYPE	GPU COUNT	GPU AVAIL	CPU AVAIL	GB MEM AVAIL
ac064	a30	2	**1**	88	456
ac013	pvc	4	**4**	96	488
ac024	pvc	8	**8**	96	488
ac100	pvc	2	**2**	96	488
ac101	pvc	4	**4**	96	488
ac082	pvc	2	**2**	96	488
ac085	pvc	4	**4**	96	488
ac099	pvc	4	**4**	96	488
ac103	pvc	2	**2**	96	488

gpuavail utilizes code from the ood-example-ps github repo which is an app that displays the output of the `ps` command



https://github.com/OSC/ood-example-ps

All necessary files including the gpuavail script and OOD Passenger App files are available in the github.com tamu-edu repository



https://github.com/tamu-edu/dor-hprc-ood-apps/tree/main/gpuavail

gpuavail OOD App Installation Requirements

- The gpuavial script reads the output from the scontrol show node command which requires the slurm.conf to be configured using the Gres parameter

- Slurm configuration file slurm.conf must use Gres configuration parameter

```
NodeName=ac[049,055,096]
    CPUs=96 Boards=1
    SocketsPerBoard=2
    CoresPerSocket=48
    ThreadsPerCore=1
    RealMemory=500000
    Gres=gpu:h100:4
    ...
```

gpuavail script also works as a command line utility

```
[userid@aces ~]$ gpuavail

     CONFIGURATION
 NODE                NODE
 TYPE                COUNT
 ---------------------
 gpu:pvc:4           17
 gpu:pvc:2           11
 gpu:pvc:8           4
 gpu:h100:4          3
 gpu:h100:8          2
 gpu:a30:2           1
 gpu:h100:2          1

          AVAILABILITY
 NODE      GPU   GPU   GPUs  CPUs  GB MEM
 NAME      TYPE  COUNT AVAIL AVAIL AVAIL
 --------------------------------------------
 ac064     a30   2     1     88    456
 ac082     pvc   2     2     96    488
 ac101     pvc   4     4     96    488
 ac086     pvc   2     2     96    488
 ac097     pvc   2     2     96    488
 ac039     pvc   4     4     96    488
 ac081     pvc   4     4     96    488
 ac089     pvc   4     4     96    488
 ac011     pvc   4     4     96    488
 ac023     pvc   4     4     96    488
```

Michael Dickens

Research Scientist

High Performance Research Computing

Global Open OnDemand Conference

HARVARD UNIVERSITY | MARCH 18–20, 2025

gpuavail app github repo

https://github.com/tamu-edu/dor-hprc-ood-apps/tree/main/gpuavail